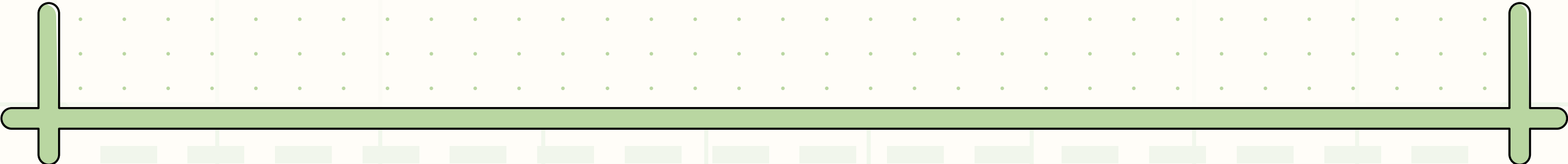
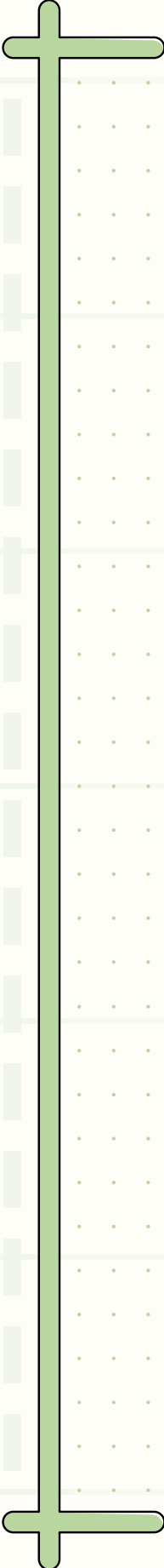


INTRODUCTION TO TECHNICAL SCIENTIFIC WRITING

by Kseniia Harshina

17.01.2025





WHAT IS SCIENTIFIC WRITING?

✦ WHAT IS SCIENTIFIC WRITING? ✦



Scientific writing is a subset of academic writing that focuses specifically on **reporting empirical research, experiments, and studies** in a clear, reproducible, and structured manner.

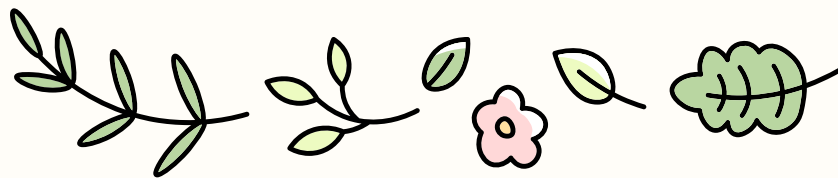


DIFFERENCES IN WRITING STYLES

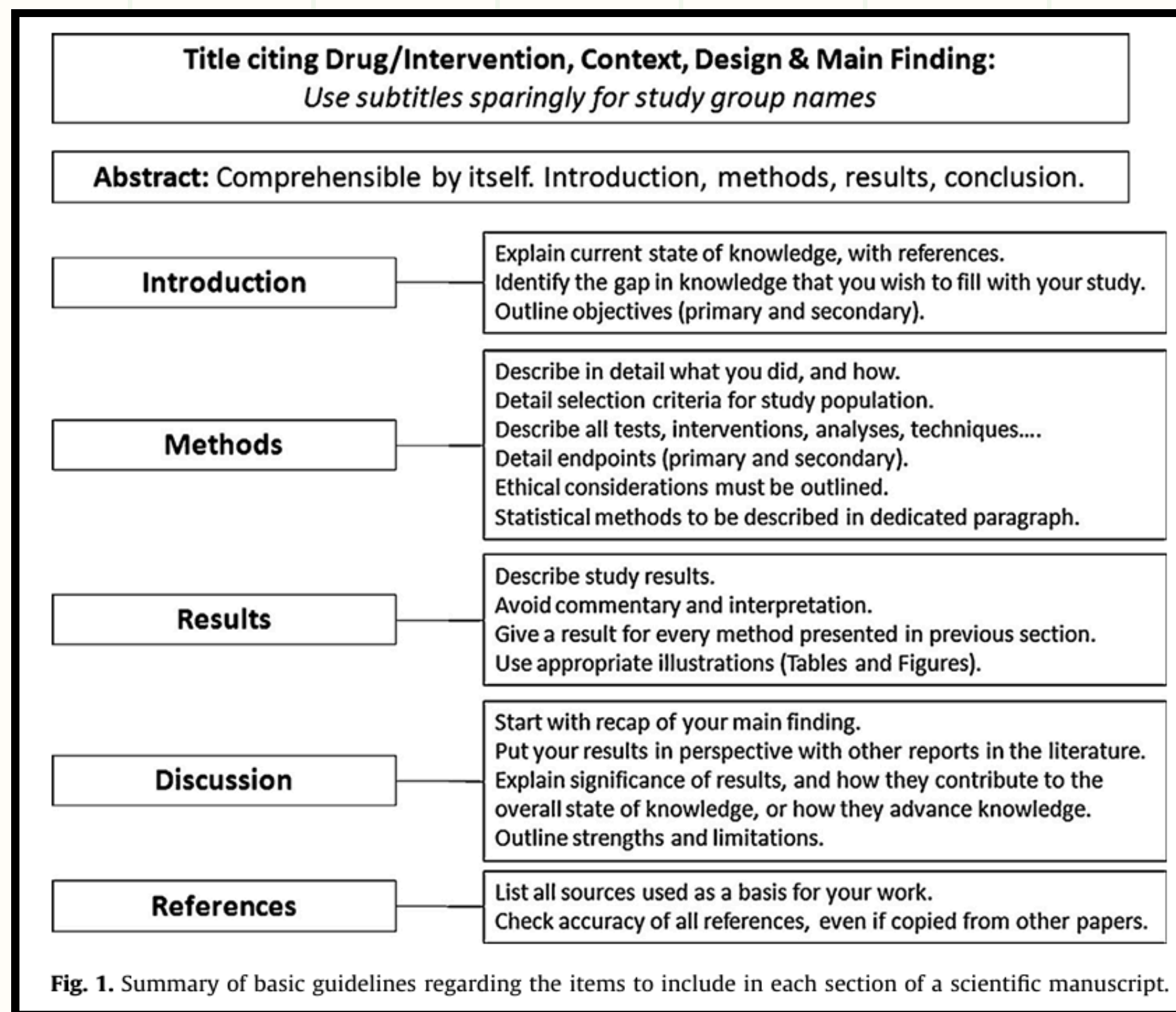
DIFFERENCES IN WRITING STYLES

Aspect	Scientific Writing	Academic Writing
Purpose	Communicate research findings concisely.	Broader focus: essays, critical analysis, theories.
Audience	Specialists in a specific field.	Academics across various disciplines.
Structure	Highly structured (IMRAD format).	Flexible depending on the discipline.
Tone	Objective, factual, precise.	Can include subjective interpretations.
Use of Data	Focused on data-driven evidence.	May include interpretive or theoretical content.

KEY GUIDELINES FOR SCIENTIFIC WRITING



KEY GUIDELINES FOR SCIENTIFIC WRITING



Writing a scientific article: A step-by-step guide for beginners

F. Ecarnot*, M.-F. Seronde, R. Chopard, F. Schiele, N. Meneveau

EA3920, Department of Cardiology, University Hospital Jean-Minjoz, 3, Boulevard Fleming, 25000 Besançon, France

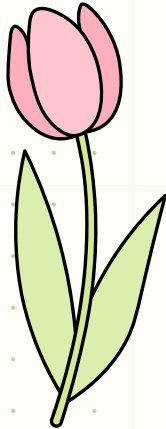
✧ KEY GUIDELINES FOR SCIENTIFIC WRITING ✧

Preparation

- 
- Conduct a literature review and organize references.
 - Choose a target journal and tailor your writing accordingly.
 - Understand the audience and formatting requirements.



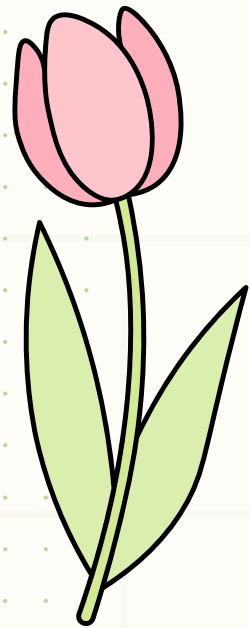
Follow the IMRAD Format

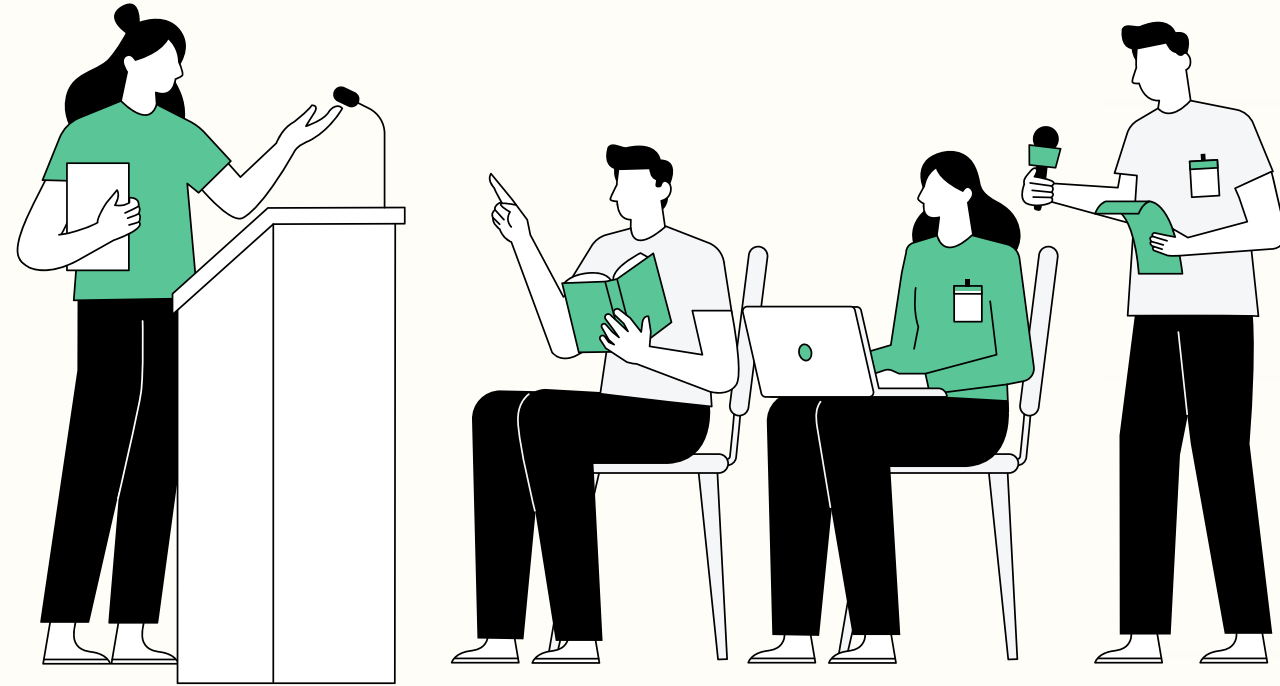
- 
- **Introduction:** State the background, knowledge gap, and objectives.
 - **Methods:** Detail study design and analysis; ensure reproducibility.
 - **Results:** Present findings clearly with figures/tables; no interpretation.
 - **Discussion:** Interpret results in context, highlight significance, and address limitations.
 - **Abstract:** Concisely summarize the entire study.
 - **Title:** Include keywords and keep it clear and specific.

✧ KEY GUIDELINES FOR SCIENTIFIC WRITING ✧

Tools

- Overleaf / LaTeX for collaborative writing
- The AI question
- Open-source code or Reproducibility (where applicable)
- Know your target audience and potential reviewer background: some communities are interdisciplinary, not just tech
- Prepare an exciting presentation!
- You can already start writing and submitting papers during your Masters or Bachelors
- There are usually many different tracks in conferences, look for the one that is applicable to your study

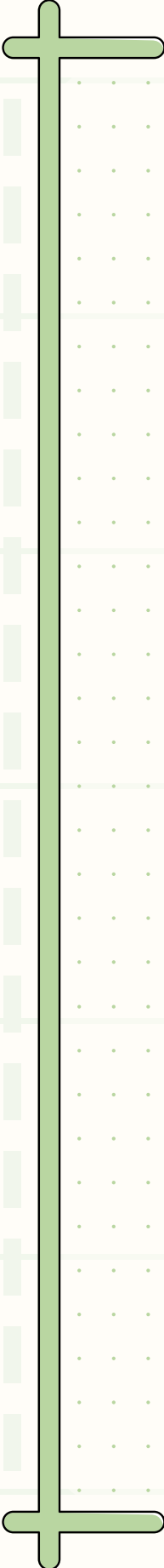




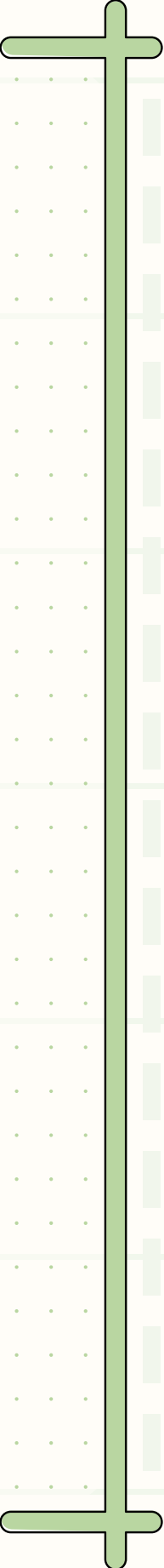
UNDERSTANDING CONFERENCE SUBMISSIONS

✦ UNDERSTANDING CONFERENCE SUBMISSIONS ✦

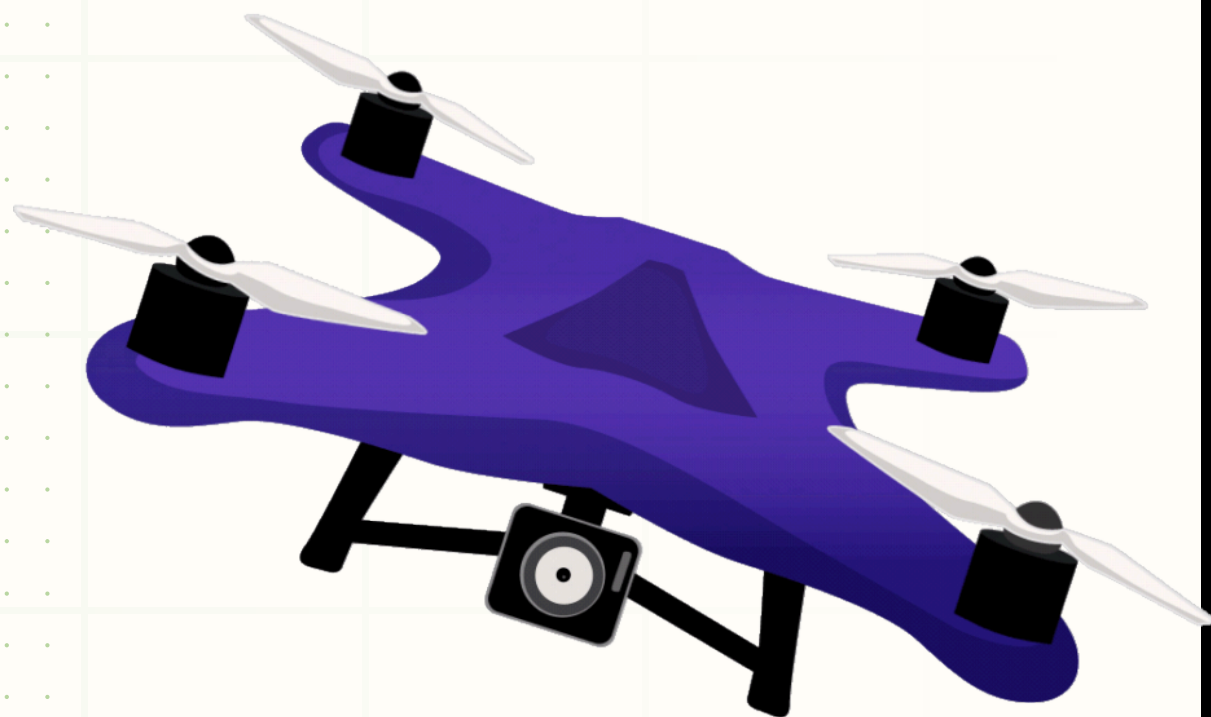




ANALYZING SCIENTIFIC PAPERS



ANALYZING SCIENTIFIC PAPERS



Reinforcement Learning-based Countermeasures against Attacking UAV Swarms

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Abstract—The development of autonomous swarm behavior for UAV swarms has increased significantly in recent years. Many applications such as collective transport, exploration of unknown territory or target search and delivery benefit from the flexibility, scalability and robustness of the swarm approach. Besides new application possibilities, these characteristics might also be used for malicious or dangerous purposes like autonomous target-oriented attacks. To date, research is lacking intelligent countermeasures to intervene in attacking UAV swarms. Typical defense mechanisms employ attack-defense confrontation which increases the risk of collateral damage as drones might fall from the sky. Rather than creating a confrontation, we focus on developing countermeasures to intelligently mislead or delay attacks on a target. Therefore, we explore two multi-agent deep reinforcement learning strategies for defender UAVs to intervene in target-oriented attacks of intelligent UAV swarms. Both strategies are based on the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm and aim at preventing or at least delaying attacks. Via simulations we model and evaluate the performance of both methods and compare it to a baseline approach.

Index Terms—attacking UAV swarms, swarm intelligence, swarm attack, drone defense, anti-drone system, reinforcement learning

As machine learning (ML) has been able to solve a variety of problems from countless fields in recent years, we want to explore ML approaches to create novel countermeasures against attacking UAV swarms. In this paper, we consider for the first time in literature a ML-based attacking UAV swarm, and we propose to intervene or delay its attack by inducing defender UAVs. To do so, we consider to use Reinforcement Learning (RL) approaches on both sides, for attackers and defenders. RL is an intuitive approach to the task of intervening in an attacking swarm as it can be easily presented as a game: there are two opposing teams (attacking and defending UAVs) with clearly established goals and winning conditions. For the purpose of keeping the attackers away from the target, we explore and evaluate two different multi-agent deep RL strategies, which are both based on the multi-agent deep deterministic policy gradient (MADDPG) algorithm [4]. The first strategy is based on communication, and thus assumes that defenders have access to the communication channel of attackers (*communication-based defense*). To avoid this requirement, we introduce a second strategy, which aims at physical protection of the target (*dynamic shield defense*). Both

FORMATTING GUIDELINES: IEEE

✦ IMRAD Format:

- Introduction, Methods, Results, and Discussion structure.

✦ Audience Consideration

- Written for a technical audience familiar with the field of reinforcement learning and UAVs.

✦ Tables and Graphs:

- Visuals are heavily used to present quantitative data effectively.

✦ Objective Tone:

- Results are reported without interpretation in the Results section.

ANALYZING SCIENTIFIC PAPERS



Behind the Screens: An Exploratory Study on Gamer Types and Identities in Game Jams

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ABSTRACT

Game jams are events where development teams collaborate to create games in days or weeks, sharing common themes and limitations. However, it is unclear how participants in game jams fit into the larger gaming culture regarding their motivations and identities as *gamers*. Our paper surveys two game jam populations to determine their player motivation profiles using the Hexad-12 questionnaire, roles in the development teams at a game jam, their correspondence to the *gamer* and *LGBTQ+* identity, and their playtime per week. The results show that the population in our samples does not differ significantly from the one reported in the original Hexad study. We found correlations between the Hexad player types and the development roles at a game jam. People interested in Game/Level Design and people reporting having fun as a reason for attending the game jam were more likely to be Achiever types, people interested in Art were more likely, while those interested in Programming were less likely to be Disruptor types, and the more participants identified as a gamer, the more likely they were Player types, and/or preferred the Game/Level Design role. Furthermore, we found that both samples identified to some extent with the term *gamer*. Additionally, the more participants leaned towards identifying as a *gamer*, the more likely they were Players and/or preferred the Game/Level Design hat.

Symposium on Computer-Human Interaction in Play (CHI PLAY Companion '23), October 10–13, 2023, Stratford, ON, Canada. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3573382.3616086>

1 INTRODUCTION

In the last two decades, game jams have become an increasingly popular phenomenon that revolve around the creation of video games. A game jam is an "[...] accelerated opportunistic game creation event where a game is created in a relatively short time frame exploring given design constraint(s) and end results are shared publically" [16]. They are closely connected to video game and development cultures, and attract participants who identify with gaming culture to some extent. Moreover, game jams can differ significantly from one another regarding their scale, settings, mode, and target population. The experiences of participants can therefore also differ greatly and are highly individual.

While game jams have a history of being tech- and programmer-focused [18], they have become more accessible toward less tech-savvy participants throughout the past two decades [7]. However, it is frequently noted that game jams are predominantly attended by people with similar identities [25] [6] [9], namely white, young, cis men, an identity that is also predominant in the game industry [1]. In contrast to this, as organizers of game jams, we have noticed

FORMATTING GUIDELINES: ACM

- ✦ Language:
 - Tailored for a multidisciplinary audience familiar with HCI concepts.
- ✦ Work-in-Progress Style:
 - The paper reports preliminary findings.
 - future research directions and limitations are explicitly outlined.
- ✦ Objective Tone:
 - Avoids subjective or speculative statements.
 - focuses on presenting empirical data and logical interpretations.

PROBLEMS WITH SCIENTIFIC WRITING

The myth of objectivity

The lack of a rigorous review process leads to oversaturation of subpart research papers

Quantity over Quality

Lack of Diversify and Accessibility

Lack of Interdisciplinarity

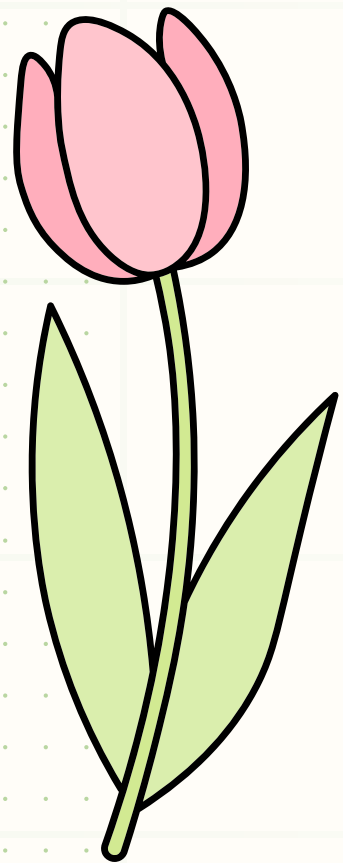
Lack of Science to Public skills

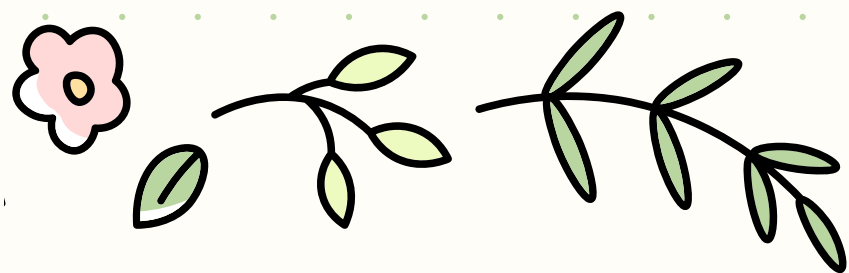
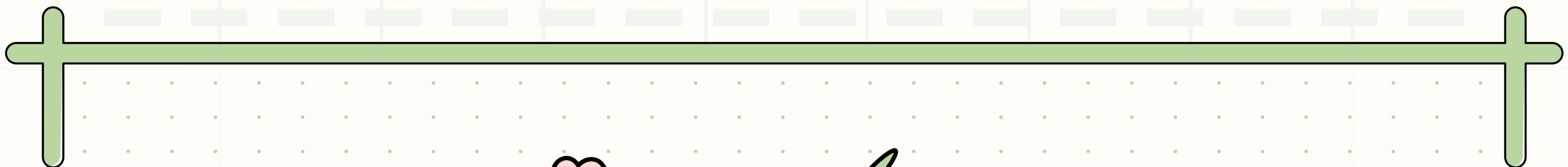
Lack of Industry/Research Collaboration

Rigid hierarchy (e.g. author order)

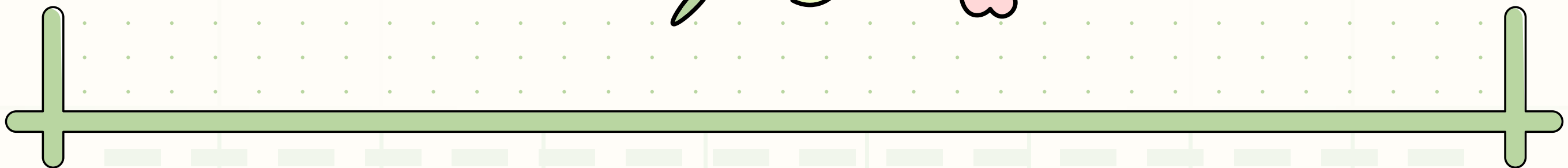
Be mindful of your role as a researcher as it relates to the participants of your studies !

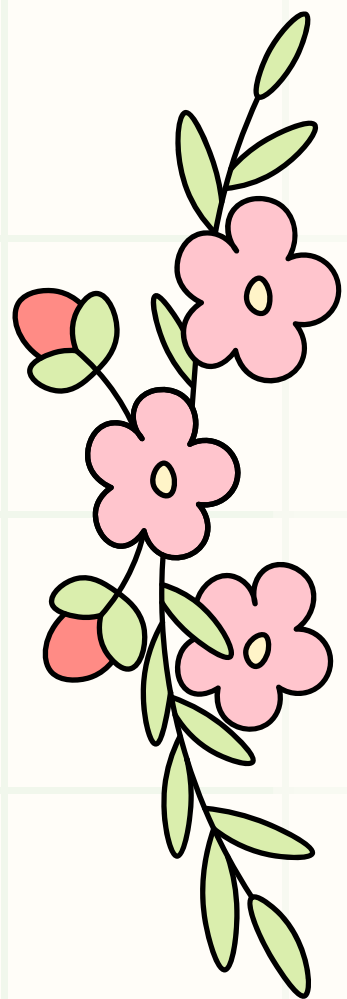
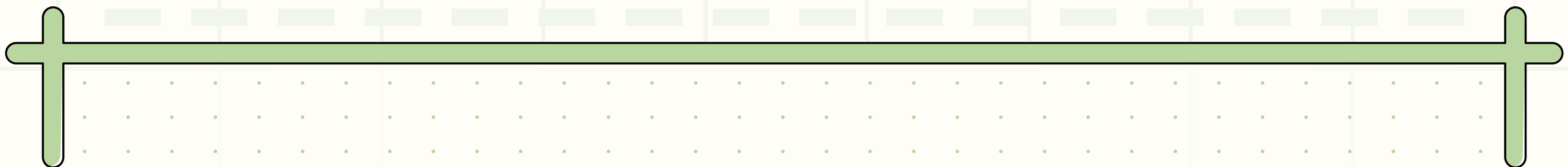
Be mindful of being in a strict research bubble





Q & A





THANK
YOU

